

Bayesian regression for language scientists

Lecture 4

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Outline

- Cross-linguistic generalizations and problems of testing them
- An overview of different solutions
- A hands-on case study with brms

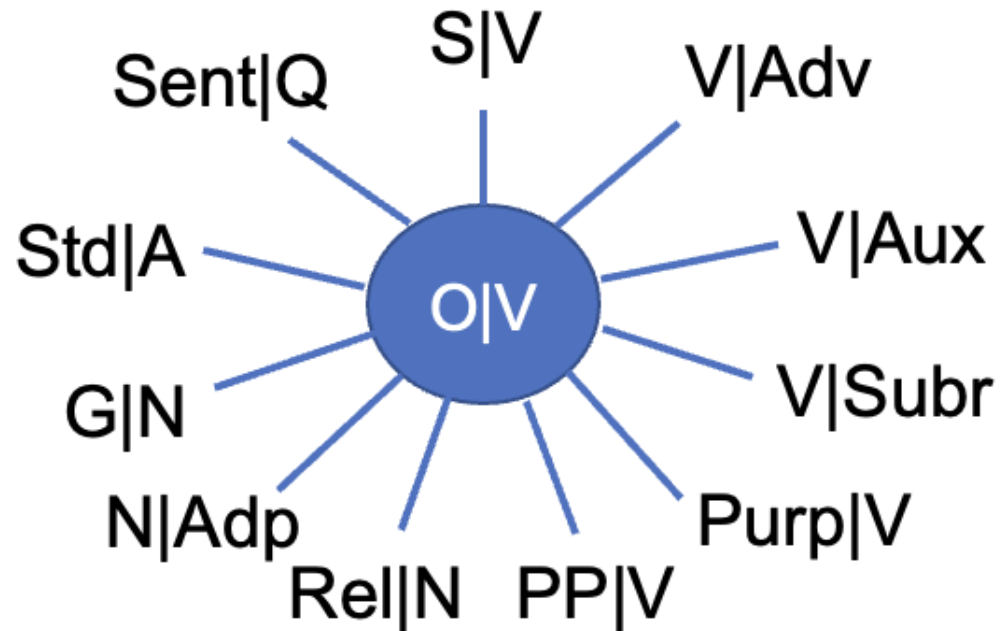
Have you heard of any language universals?

Univariate biases

- Greenberg's universal 1: "In declarative sentences with nominal subject and object, the dominant order is almost always one in which the subject precedes the object."

Subject - Object	Object - Subject
Yes (or MANY)	No (or FEW)

Correlations: Many WO universals



	Noun - Postposition	Preposition - Noun
Object - Verb	Yes	No
Verb - Object	No	Yes

Croft 2003, based on Dryer 1992

Implications: Greenberg's Universal 25

"If the pronominal object follows the verb, so does the nominal object."

	Verb - Nom. Object	Nom. Object - Verb
Pron. Object - Verb	Yes	Yes
Verb - Pron. Object	Yes	No

How to test?

- Greenberg looked at a sample of 30 languages, no statistical tests.
- Using this method, one can only test exceptionless universals (rare!)

But how to test **statistical** correlations and implications?

Traditional statistical tests

Popular in computational linguistics

Linguistic Complexity: Relationships Between Phoneme Inventory Size, Syllable Complexity, Word and Clause Length, and Population Size

Gertraud Fenk-Oczlon^{1} and Jürgen Pilz²*

Method

- "The parallel textual material used for our analysis consists of 22 simple declarative sentences encoding one proposition and using basic vocabulary. "
- "Native speakers of 61 languages from 18 language families and from all continents were asked to translate the 22 sentences into their mother tongue. Most of our informants were students, many of them linguists we met at international conferences. The basic requirement was a good knowledge of either English or German in order to be able to translate the test sentences into their mother tongue."

Language sample

Athabaskan-Eyak-Tlingit [Navajo (NAV)] **Atlantic-Congo** [Bafut (BFD), Ewondo (EWO), Lamnso (LNS), Kirundi (RUN), Yoruba (YOR)] **Austronesian** [Batak (BYA), Cham (CJA), Chuukese (CHK), Hawaiian (HAW), Javanese (JAV), Kadazan (DTB), Kemak (KEM), Malagasy (BHR), Malay (MEO), Mambae (MGM), Minangkabau (MIN), Nias (NIA), Roviana (RUG), Tagalog (TGL)] **Austroasiatic** [Vietnamese (VIE)] **Basque** [Basque (EUS)] **Chiquitano** [Chiquitano (CAX)] **Dravidian** [Telugu (TEL)] **Indo-European** [Albanian (SQI), Armenian (XCL), Bulgarian (BUL), Czech (CES), Croatian (HLV), Dutch (NLD), English (ENG), French (FRA), German (DEU), Greek (ELL), Hindi (HIN), Icelandic (ISL), Italian (ITA), Latvian (LAV), Macedonian (MKD), Norwegian (NOR), Panjabi (PAN), Persian (PER), Polish (POL), Portuguese (POR), Romanian (RON), Russian (RUS), Slovenian (SLV), Spanish (SPA), Tajik (TGK)] **Japonic** [Japanese (JPN)] **Kartvelian** [Georgian (GEO)] **Koreanic** [Korean (KOR)], **Mande** [Bambara (BAM)] **Sino-Tibetan** [Mandarin Chinese (CMN)] **Tai-Kadai** [Thai (THA)] **Turkic** [Turkish (TUR)] **Uralic** [Estonian (EKK), Finnish (FIN), Hungarian (HUN)] **Uto-Aztecan** [Hopi (HOP)] **Western Daly** [Maranunggu (ZMR)].

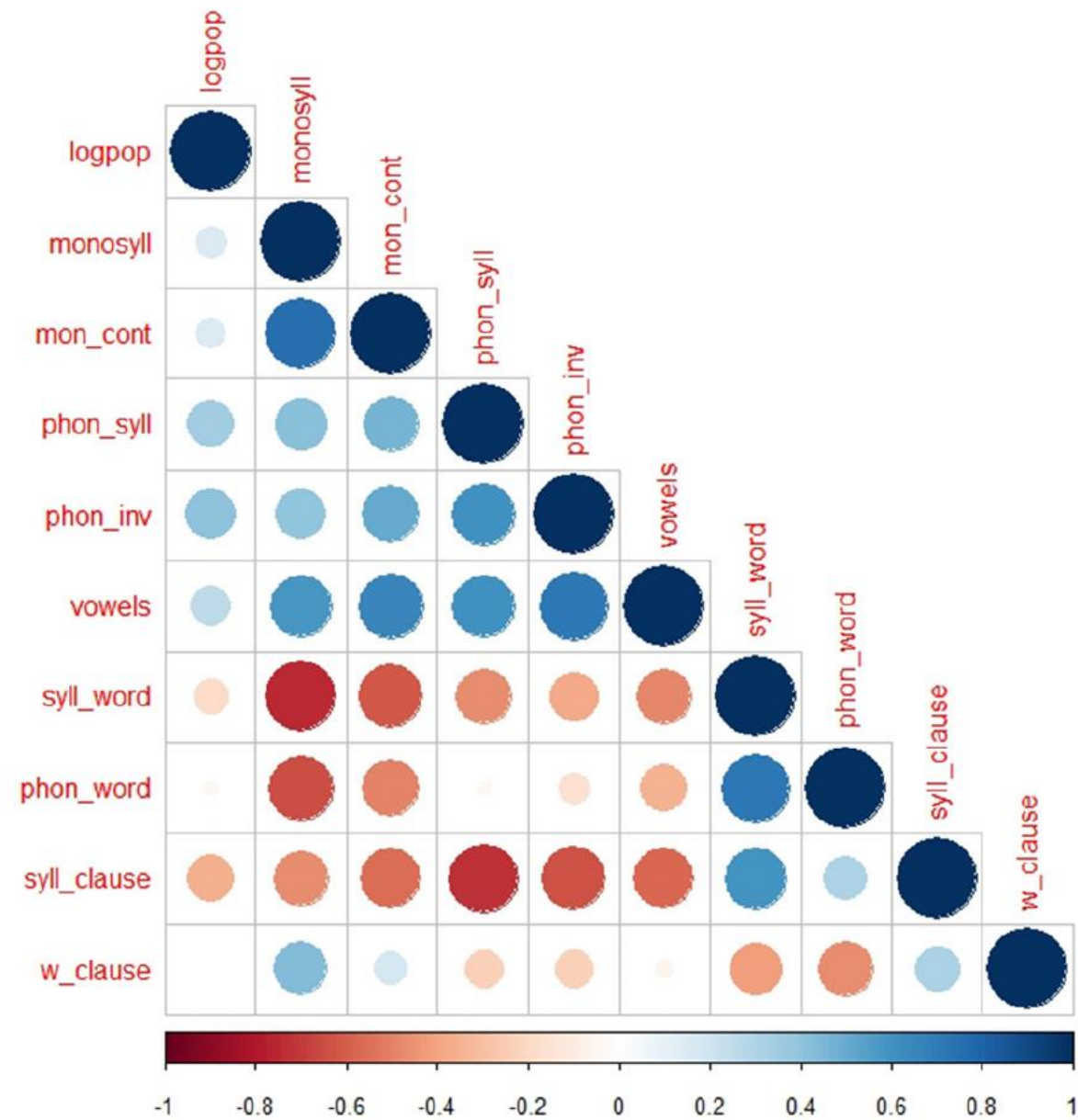


FIGURE 2 | Correlation matrix between the linguistic variables and log of population sizes.

Galton's Problem

- Sir Francis Galton (comparative anthropology): similarity between observations correlates with their **proximity in evolutionary time**
- If we don't control for this non-independence, significance can be inflated.
- Plus the problem of similarities due to language and cultural **contact**

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What can we do?

- Make the languages in our sample as independent as possible
- Aggregate data by genealogical and geographic units
- Incorporate the dependencies in the random effect structure of mixed-effects regression models

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Independent languages

- Perkins (1980, 1988) and Bybee (1985) – a sample of 50 languages:
 - no more than 1 language per language phylum
 - no more than 1 language per known cultural and geographic area
- Strengths: intuitively understandable, no need for complex statistical tests
- Problems:
 - difficult to find data, we can end up testing the same set of languages
 - usually results in a relatively small sample
 - what is the right level of analysis (areas, genealogy)?

If we wanted to test a language universal relying only on our knowledge,

- How many languages could we gather data about in this room?
- How many **independent** data points would we be able to collect?

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Dryer's approach: genera and macroareas

	AFRICA	EURASIA	SEASIA&Oc	AUS-NEWGUI	NAMER	SAMER	TOTAL
OV&Postp	15	26	5	17	25	19	107
OV&Prep	3	3	0	1	0	0	7
VO&Postp	4	1	0	0	3	4	12
VO&Prep	16	8	15	6	20	5	70

TABLE 1. Adposition type.

Key: The numbers indicate the number of genera containing languages of the given type in the given area. The large of the two numbers for each area and for each order of verb and object is enclosed in a box. Africa includes Semitic languages of southwest Asia; Eurasia = Europe and Asia, except for southeast Asia, as defined immediately; SEAsia&Oc = Southeast Asia (Sino-Tibetan, Thai, and Mon-Khmer) and Oceania (Austronesian); Aus-NewGui = Australia and New Guinea, excluding Austronesian languages of New Guinea; NAmer = North America, including languages of Mexico, as well as Mayan and Aztecan languages in Central America; SAmer = South America, including languages in Central America except Mayan and Aztecan languages.

From Dryer 1992

Random sampling/bootstrapping

- Levshina (2021): correlations between different corpus-driven features related to A and P
- a sample of 30 languages, mostly Indo-European
- correlation analysis:
 - 1000 random samples (one language per genus)
 - Spearman correlation coefficients are logged and then averaged
 - Permutation for null hypothesis significance testing

	Case Marking	Tight Semantics	Rigid Word Order	Verb between Subj and Obj
Case Marking		0.49 0.44	-0.67 -0.66	-0.47 -0.22
Tight Semantics	0.49 0.44		-0.16 0.26	-0.44 -0.27
Rigid Word Order	-0.67 -0.66	-0.16 0.26		0.28 0.01
Verb between Subj and Obj	-0.47 -0.22	-0.44 -0.27	0.28 0.01	

Pros and cons

- Strengths:
 - We can include as many languages as we like
- Problems:
 - What is the unit of analysis?
 - "The groups identified as genera are intended to be maximal groups with a time depth no greater than 4000 years. Because our current knowledge about the time depths of most genetic groups is rather meagre, considerable guesswork has been involved in identifying these genera." (Dryer 1992: fn2)
 - Separating areas can be tricky: "The choice of areas and where to draw their boundaries is somewhat arbitrary. (..._ The crucial question, however, is to what extent the results discussed here would have been different had a different set of areas been chosen." (Dryer 1992: fn3)
 - Family/genus size doesn't matter: Very big language families on a par with isolates
 - Low number of data points due to aggregation, loss of data

What can we do?

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Mixed models

- The de facto standard in psycholinguistics and corpus-based studies of language variation
 - repeated measurements (subjects, stimuli)
 - hierarchical/nested data (sentence < text < author in a corpus)
- Mixed models combine:
 - Fixed effects: overall relationships shared across all observations
 - Linguistic variables and features we are interested in
 - Random effects: group-specific variation (e.g., participants, schools, sites, stimuli in experiments)
 - Genealogical and geographic information

Differential Object Marking (DOM)

Spanish

- Inanimate: no case markers

Ve *la* *casa*.
see.PRS.1SG DEF house
"I see the house."

- Animate and specific: case marker *a*

Ve *a* *la* *mujer*.
see.PRS.1SG OBJ DEF woman
"I see the woman."

Referential scales in DOM

Human > Animal > Inanimate

Definite > (Indefinite) Specific > Non-specific

Given > New

Pronoun > Noun



(MORE) MARKING OF P

NO (LESS) MARKING OF P

Silverstein (1976), Bossong (1991: 159), Comrie (1986: 94), Croft (2003: 132), Haspelmath (2021), Schmidtke-Bode & Levshina (2018); but Bickel et al. (2015)

Correlations between rank and marking

Table 15: Data coding for the rank-based GLM (segment)

MContinent	Family	System	RefCat	Number	Marking	Rank
Africa	Adamawa-Ubangi	Gbeya	1	SG	marked	12
Africa	Adamawa-Ubangi	Gbeya	1	NSG	marked	12
Africa	Adamawa-Ubangi	Gbeya	2	SG	marked	12
Africa	Adamawa-Ubangi	Gbeya	2	NSG	marked	12
Africa	Adamawa-Ubangi	Gbeya	3	SG	marked	3
Africa	Adamawa-Ubangi	Gbeya	3	NSG	marked	3
Africa	Adamawa-Ubangi	Gbeya	N _{high}	SG	unmarked	N _{high}
Africa	Adamawa-Ubangi	Gbeya	N _{high}	NSG	unmarked	N _{high}
Africa	Adamawa-Ubangi	Gbeya	N _{low}	SG	unmarked	N _{low}
Africa	Adamawa-Ubangi	Gbeya	N _{low}	NSG	unmarked	N _{low}

Data from AUTOTYP (Bickel et al. 2023)
<https://doi.org/10.5281/zenodo.7976754>

Schmidtke-Bode & Levshina 2018

Multilevel logistic model

- Response variable: marked or unmarked P (object)
- Fixed effects:
 - Rank on the referential hierarchy
 - 1st or 2nd person > 3P pronouns > High-ranking noun (animate, definite, given...)
>Low-ranking noun (inanimate, indefinite, new...)
 - Number
 - Macrocontinent
 - and interactions between those
- Random intercepts: "System" (language) and Family

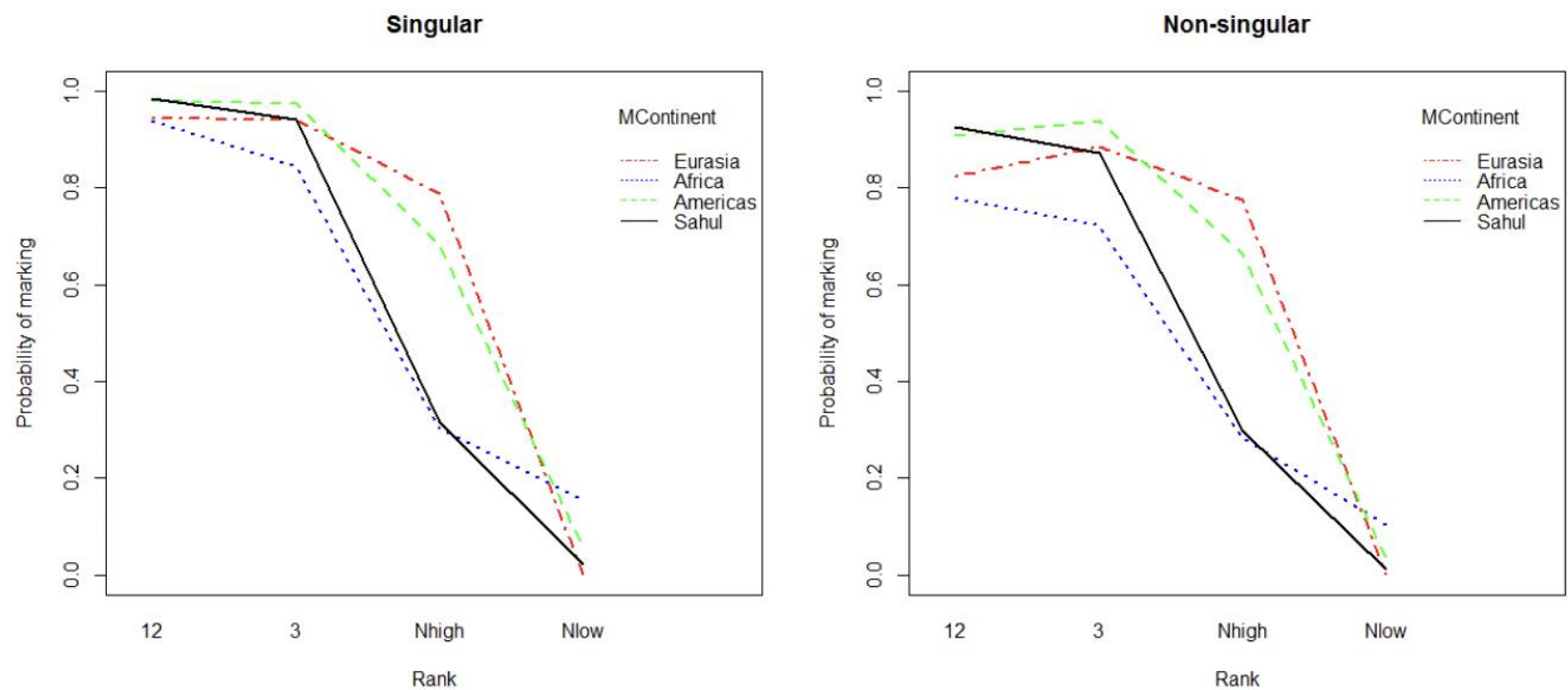
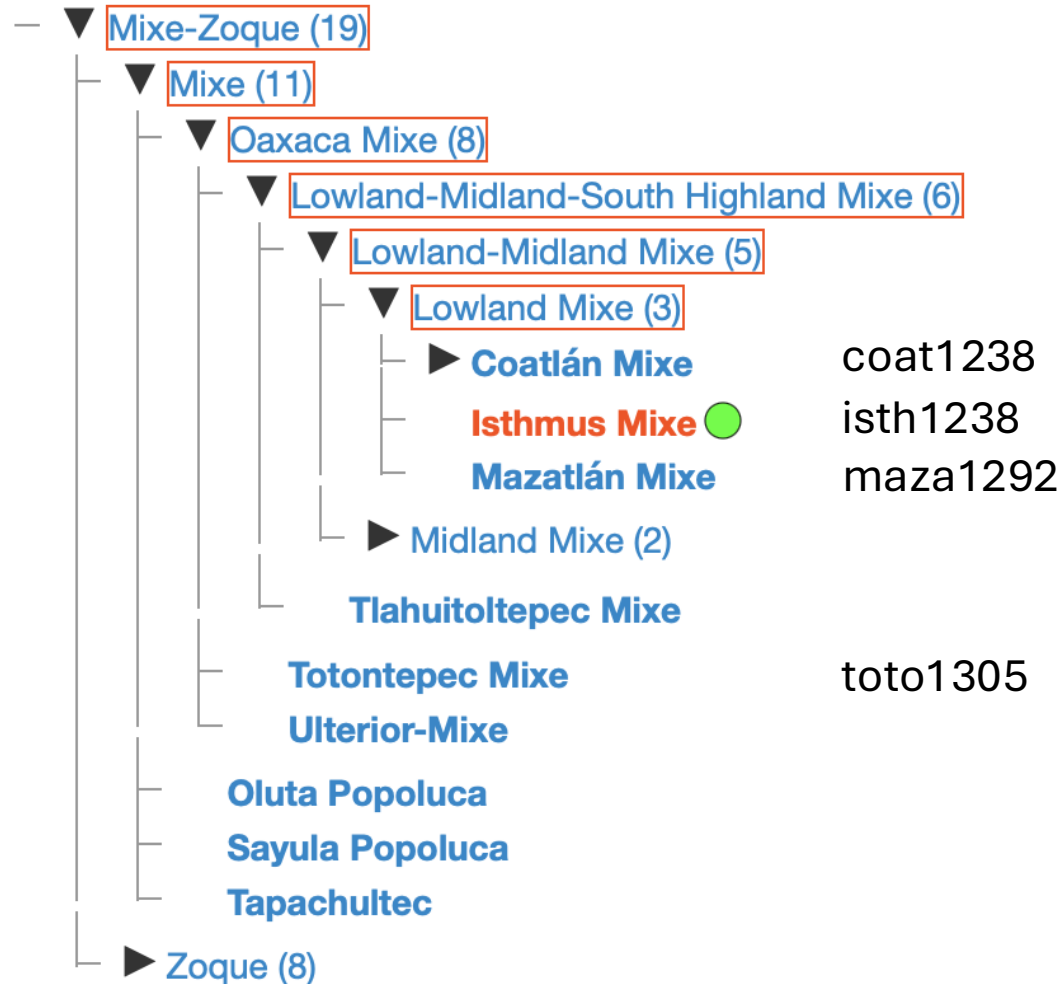


Figure 1: Influence of ranks on the probability of marked Ps in the singular (left) and the non-singular (left), by macro continent.

Pros and cons of using groups as random effects

- Strengths:
 - We can use all languages as data points, no loss of data
 - A well-established statistical method
 - Flexibility: all kinds of variables
- Problems:
 - choosing the levels for groupings
 - loss of information from phylogenetic trees

Example: Mixe-Zoque languages



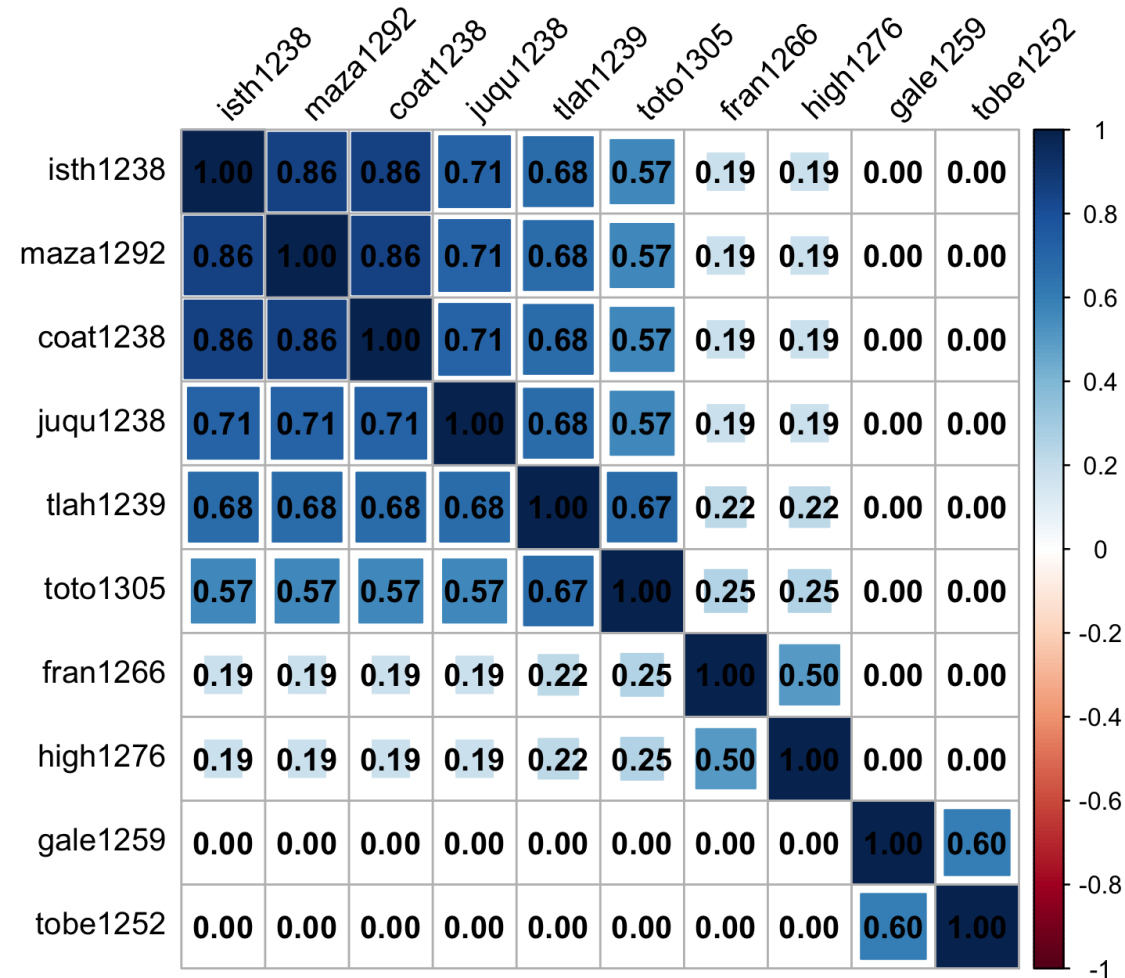
A clarification (based on a question)

- The assumption of independent data points can also be accounted for when using an lme4 mixed-model.
- There is no difference between lme4 and brms in modelling data with language genera, families, macrocontinents as groups (except for normal Bayesian 'perks')
- But there is one cutting-edge approach in which brms shines!

Hierarchical phylogenetic regression

- Originally developed in evolutionary biology (*Hadfield & Nakagawa 2010*)
- Convenient implementation in R package *brms* (*Bürkner 2024*) – for all popular types of response variables
- Random structure reflects **correlations** between species (languoids belonging to different genera and families), based on the phylogenetic trees.

Phylolinguistic information from Glottolog

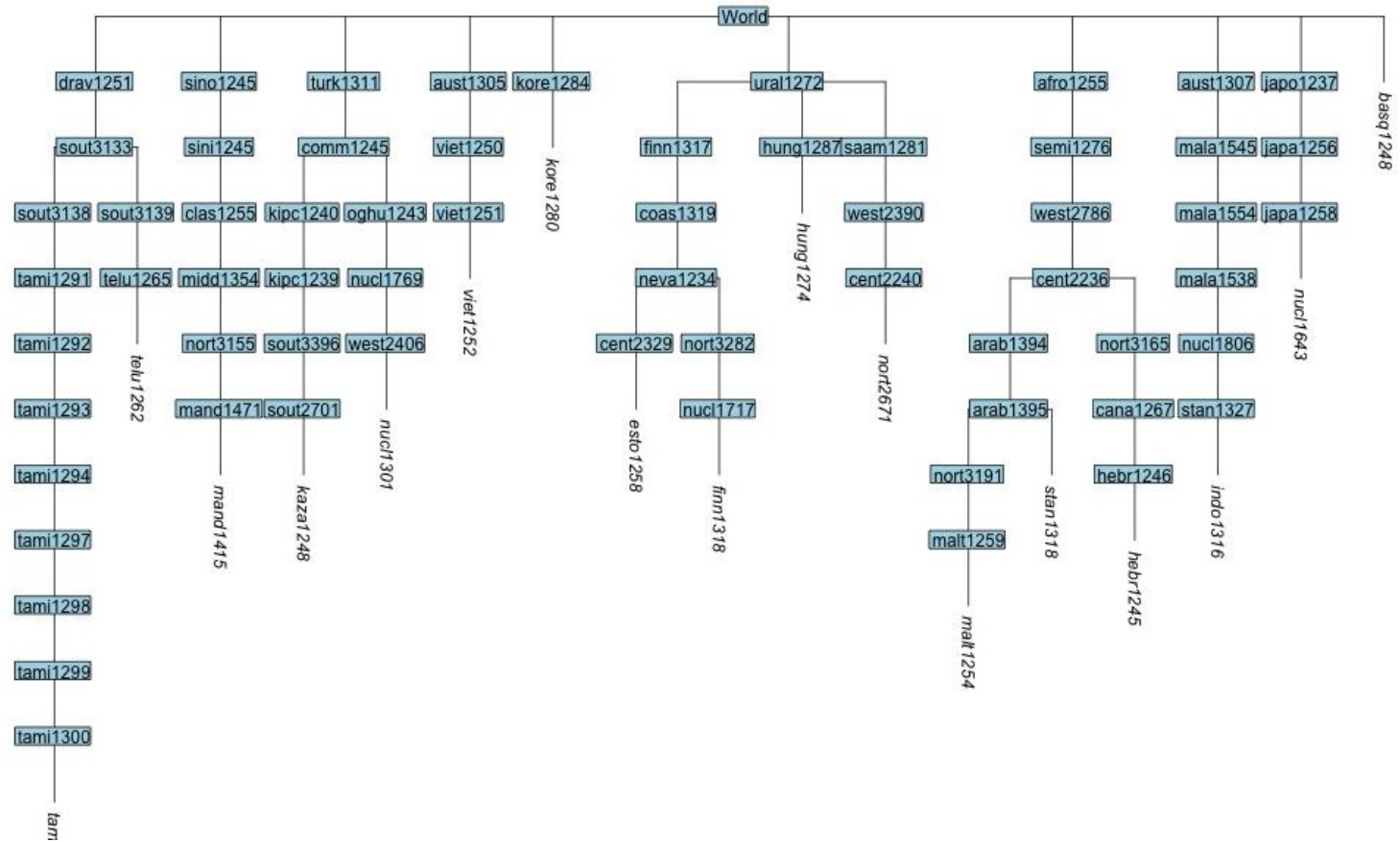


The logic

- Species that are very close together in the tree have values in the covariance matrix closer to 1, and species that are unrelated will have values 0.
- The information in the covariance matrix is added to the regression model as part of the random effects structure.
- The model assumes the estimated effects (usually intercepts) to be more similar for species which are closer together in the tree, but less so for species which are less close to each other.

One "supertree"?

- A problem: we do not know the depth at which all language families merge.
- Fortunately, we don't need to operate on a tree. We can assign zero correlations in the covariance matrix if two languages belong to different language families.



What about language contact?

- Languages that are close geographically are more likely to be in contact.
- We can use geographic coordinates to control for spatial autocorrelations
- A solution is to create a variance-covariance matrix, such that languages that are closer influence each other more.
- The covariances (correlations) can be calibrated: is diffusion likely only locally (e.g., hundreds of kilometers), or also across larger regions?

A problem with distances

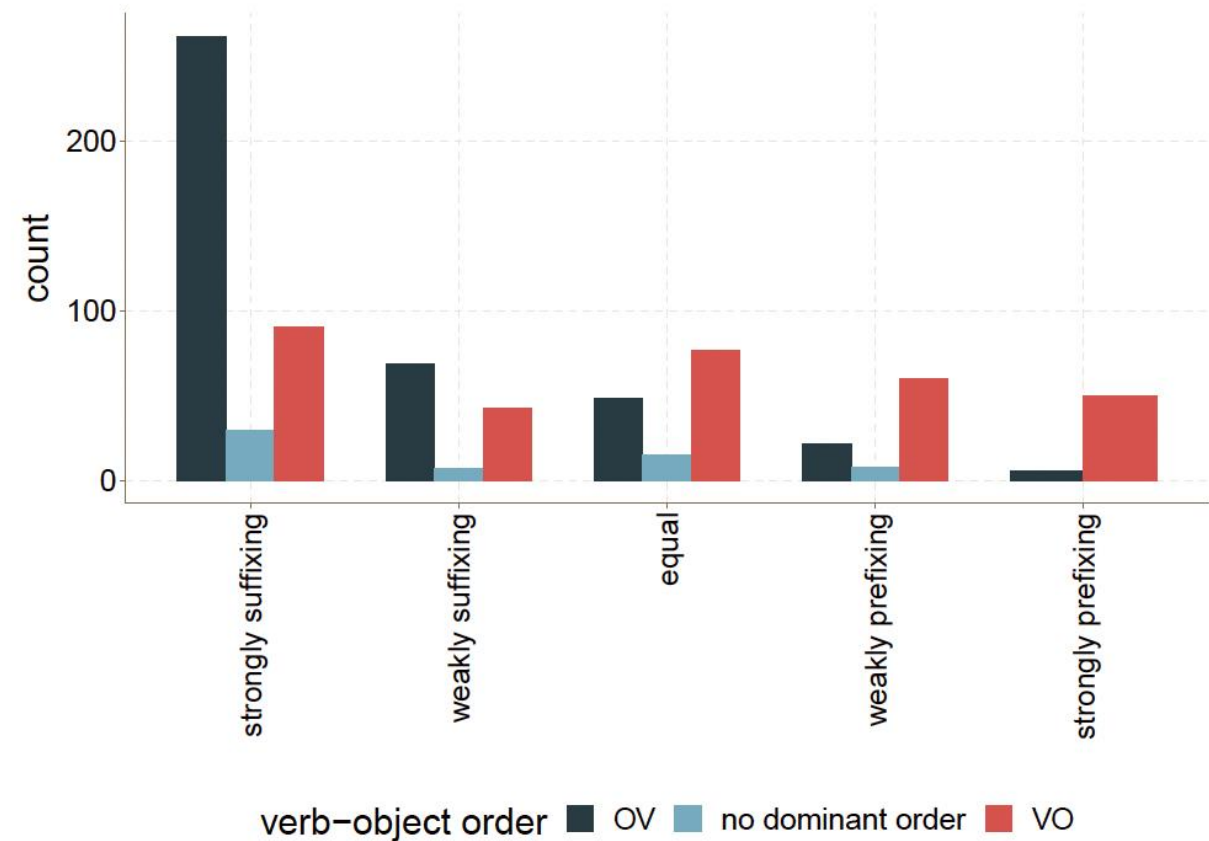
- Problem (Dryer 2018): if two languages are 100 km apart, how likely are they to display common features due to contact?
 - If in Siberia → more likely
 - If in New Guinea → less likely
- A solution is to use Gaussian Processes (GP) based on latitude and longitude (Guzmán Naranjo & Becker 2022)
 - two close languages will have a strong influence over each other
 - two observations far apart will have less influence over each other
 - the effect is inferred automatically from the data → the effect of distance can vary across areas

Pros and cons of Bayesian phylogeographic regression

- Strengths:
 - many data points
 - flexibility
 - fine-grained representations of geological and geographic information
- Problems:
 - often computationally intensive (we usually cannot test random slopes)
 - uncertainty in phylogenies (what is the best tree?). Possible solution: apply regression on a set of possible trees, incorporate the uncertainty in the regression parameters (not part of this course)
 - actual interaction between linguistic communities depends on more factors than just geographic distance between them based on 2-dimensional coordinates; coordinates are not very informative for languages spoken in a large area

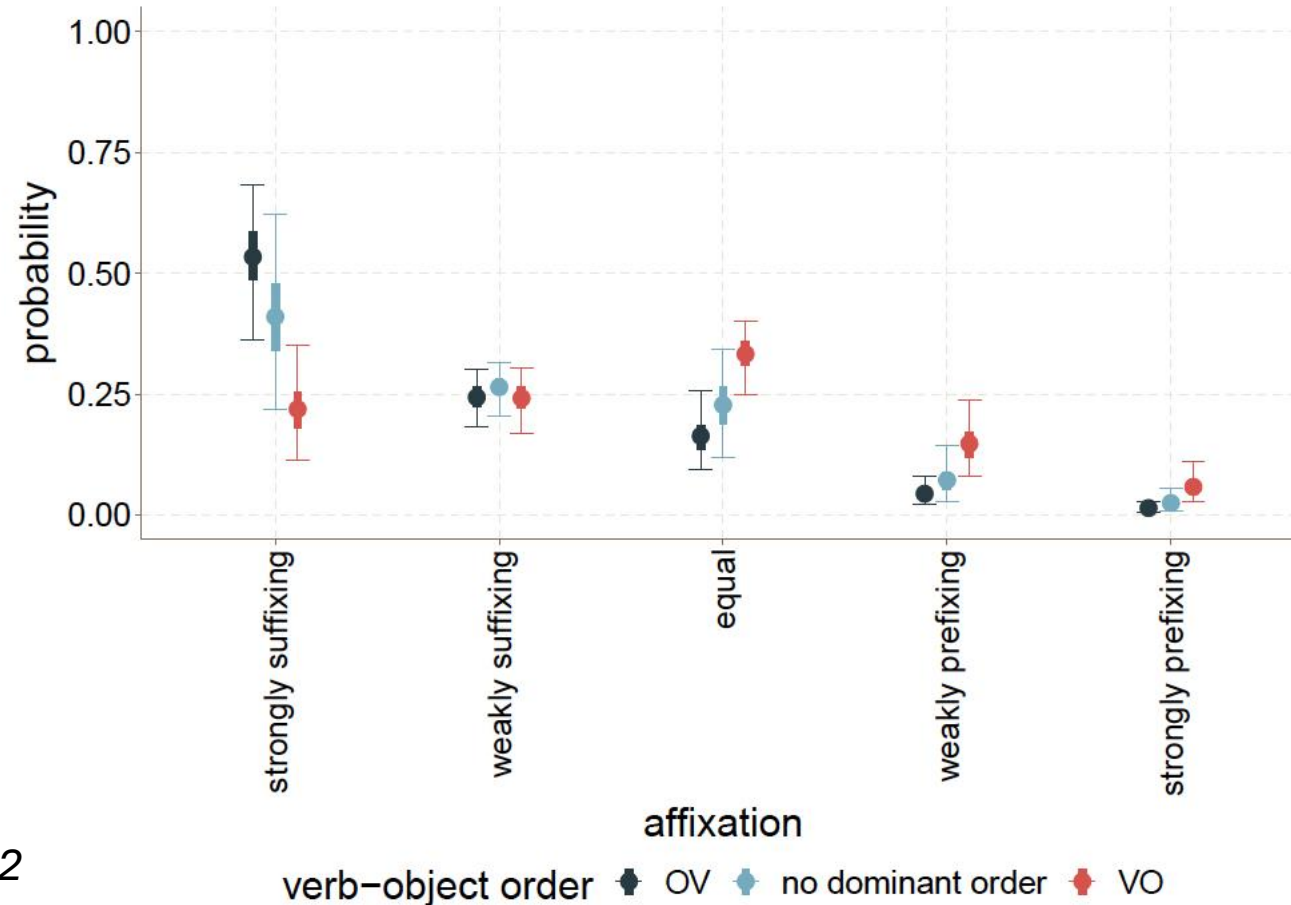
Comparison of methods

- Guzmán Naranjo & Becker (2022): a study of correlation between affix position and Object-Verb order
- Hypothesis: OV languages strongly prefer suffixation, whereas VO languages can have both.



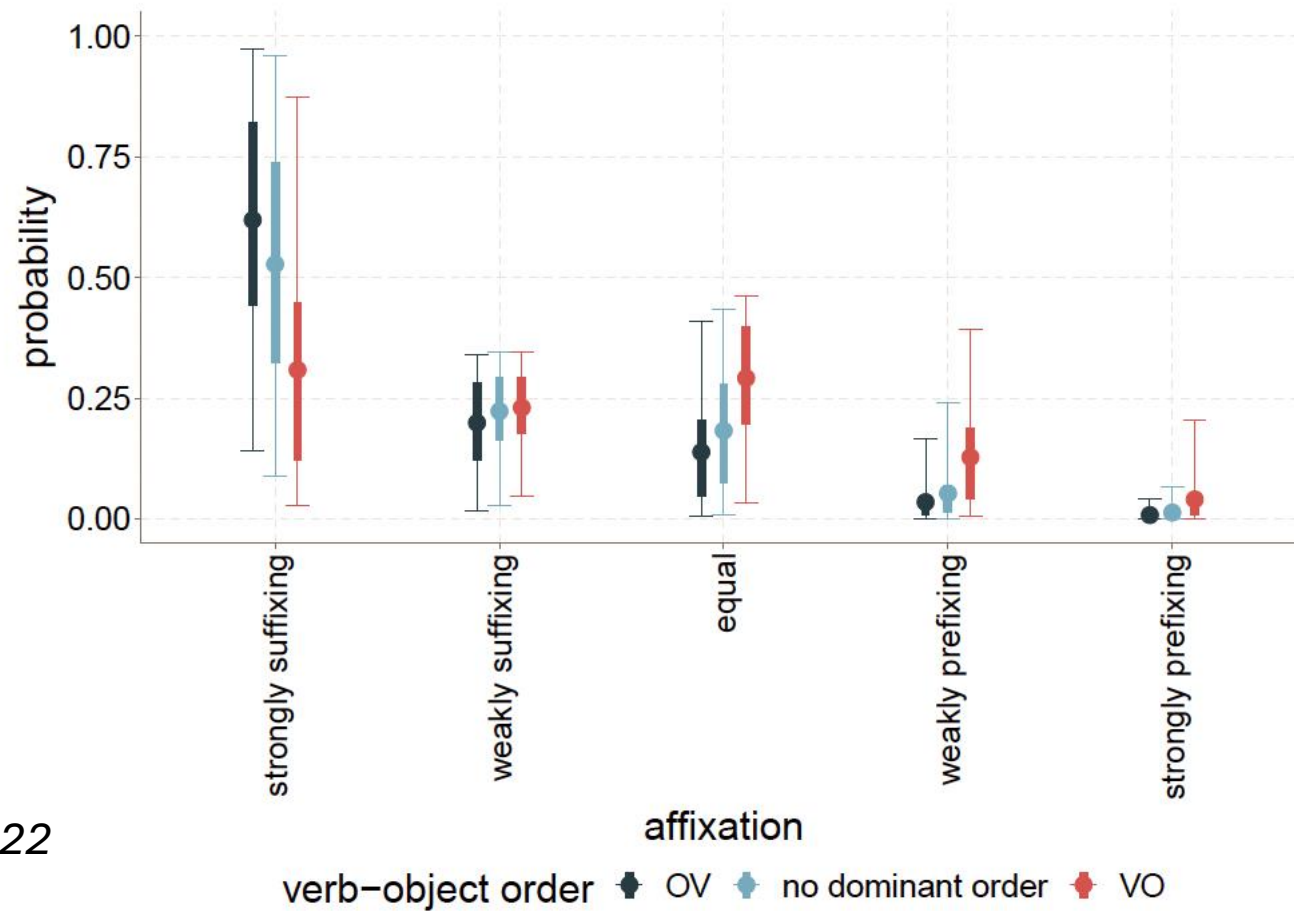
Model with "normal" random intercepts

affixation ~ VO-order + (1|family) + (1|macroarea)



Phylogeographic model

affixation ~ VO-order + micro-family covariance matrix + Gaussian Processes by macroarea

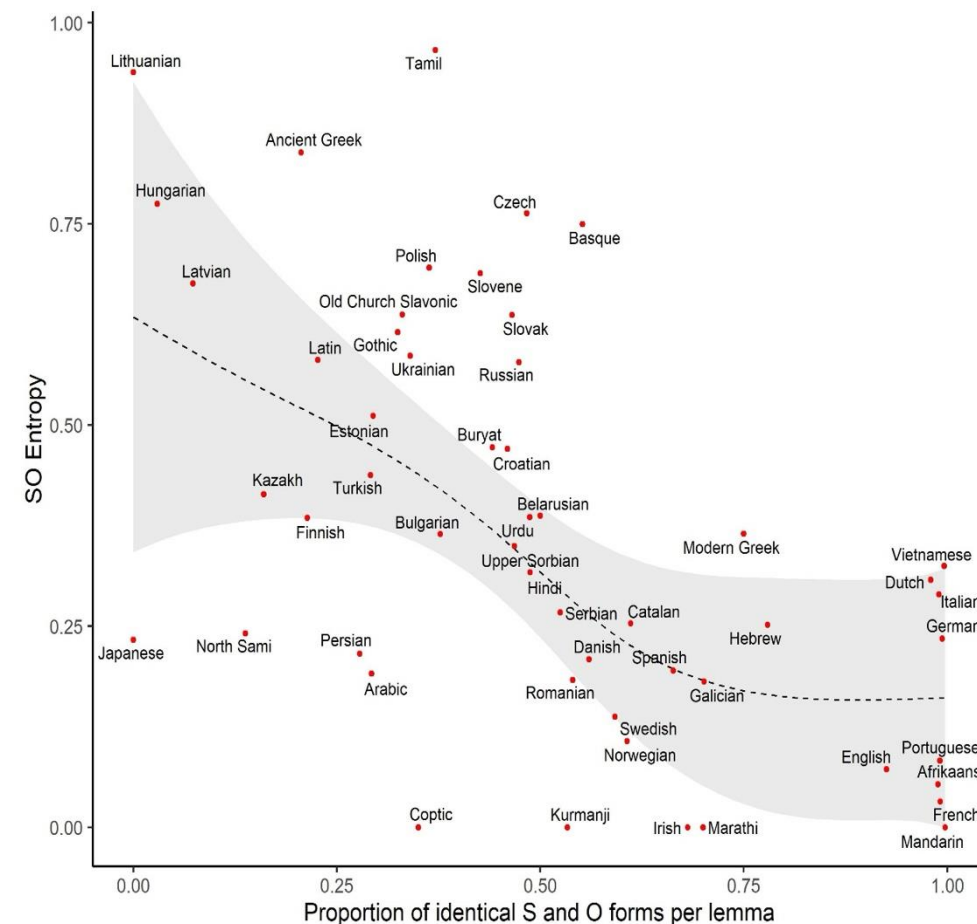


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Case study

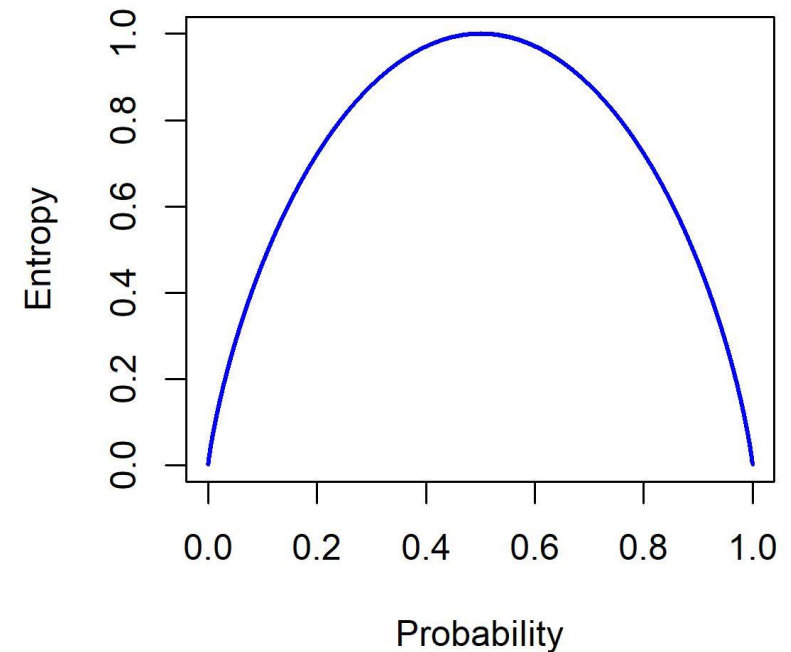
- Languages without case marking tend to have rigid word order, whereas languages with case marking allow for more flexibility.
- Levshina (2019, LingTyp): focus on S and O in Universal Dependency corpora.
- A negative correlation between the proportion of nominal lemmas that have the same S and O forms in the corpus, and entropy of SO order.



Entropy

- Proportions of nsubj + obj and obj + nsubj (only common nouns) in a transitive clause
- The higher entropy H , the greater the variability

$$H(X) = -\sum_{i=1}^2 P(x_i) \log_2 P(x_i)$$

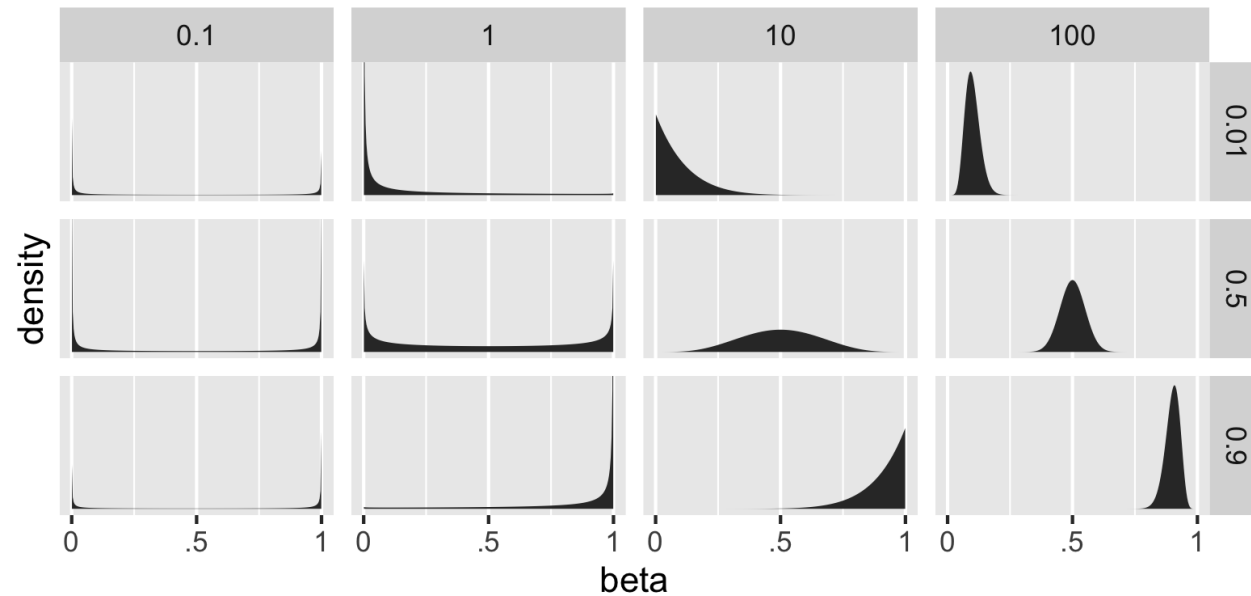


The dataset and R code can be found in the GitHub repository for the course, Day 4.

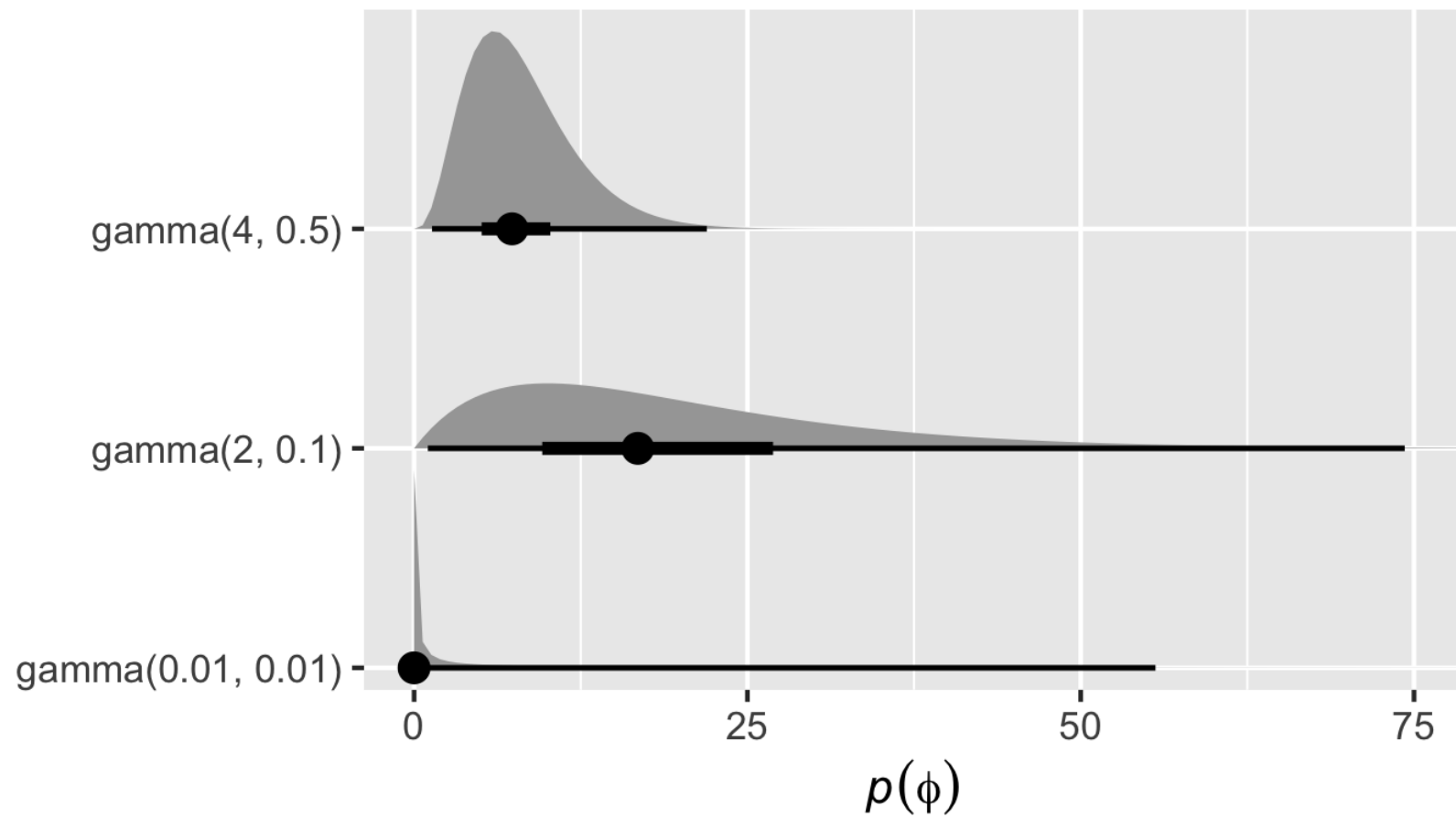
Some facts about beta regression

- The response variable is $0 < y < 1$
- The intercepts and slopes are on the logit scale (the default).
- Additional *phi* parameter that determines the shape – see columns in the plot. Beta distributions with large *phi* are similar to normal distributions.
- Default priors for *phi*: gamma (0.01, 0.01), L-shaped curve, extremely close to zero.

Beta given different μ (rows) and ϕ (columns)



Prior distributions for phi parameter



Model comparison: Technical details

- Model comparison is frequently done using the LOO criterion.
 - Ideally, we should leave out every observation and then refit the model.
 - But this is VERY impractical, so the statistics are computed using an efficient approximation.
- Some observations may produce high pareto_k values (> 0.7), which means that leaving them out affects the posteriors strongly. This can produce inaccurate LOO estimates. Causes: influential observations, model specification, random variation...
- To correct this, we can use the moment matching method as a patch. It is more computationally intensive, but still more efficient than refitting the model for every problematic observation.
- In order to be able to use this method, all parameters for inner computations which are normally discarded, should be saved by adding `save_pars = save_pars(all = TRUE)`. This increases the size of the model object.